

# Blueshift UQFF Gravitational Approach Amplifier — $\_\text{approach} = 1/(1+z)$ for Negative Redshift Systems

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## PAPER\_273: Blueshift UQFF Gravitational Approach Amplifier — $\_\text{approach} = 1/(1+z)$ for Negative Redshift Systems

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**UQFF Module:** ANDROMEDA\_UQFF\_MODULE.cpp (M31 Master Module, Session 75)

**Session:** 75 — Andromeda UQFF 2.0 Analysis

**Keywords:** blueshift, negative redshift, approach amplifier, kappa,  $z=-0.001$ , gravitational amplification, Andromeda M31, UQFF redshift coupling

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<!-- UQFF constants:  $\kappa = 5.0\text{e-}4 \text{ day}^{-1}$ ,  $[\text{SSq}] = 0.57$ ,  $M_{\text{UQFF}} = 1.43\text{e}1 \text{ TeV}$  --> ## Abstract

In all prior UQFF modules, redshift  $z \geq 0$  was assumed (receding or static systems). The Andromeda Galaxy (M31) is the nearest major galaxy and is unique among large-scale systems in having  $z = -0.001$  (blueshift — approaching the Milky Way at  $\sim 110 \text{ km/s}$ ). Applying the UQFF redshift coupling factor  $1/(1+z)$  to a system with  $z < 0$  produces  $\_\text{approach} > 1$ : a gravitational amplification factor for approaching mass systems. We define  $\_\text{approach} = 1/(1+z) = 1/0.999 = 1.001001\dots$  for M31, identifying it as the **UQFF Blueshift Gravitational Approach Amplifier**. We show that as  $z \rightarrow -1$  (hypothetical maximum approach),  $\_\text{approach} \rightarrow \infty$ , implying a **self-reinforcing merger resonance cascade**. The blueshift amplifier scales all UQFF gravitational terms simultaneously, making the total UQFF gravity of an approaching galaxy slightly but measurably stronger than a static equivalent. This is the first UQFF treatment of negative redshift as a gravitational degree of freedom.

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## 1. Introduction

Standard DPM-seeded and GR gravity treats the gravitational force between two masses as independent of relative velocity in the non-relativistic limit. Doppler blueshift is traditionally interpreted as a spectral phenomenon with no consequence for the gravitational interaction energy.

In UQFF, the redshift parameter  $z$  encodes the cosmological expansion state of the system and

enters the gravitational calculation through the factor  $(1+z)$ . For receding galaxies ( $z > 0$ ), this factor is  $> 1$  and suppresses the effective gravitational coupling. For  $z = 0$  (static), the factor is exactly 1. For  $z < 0$  (blueshift, approaching), the factor  $(1+z) < 1$ , so its reciprocal  $\_approach = 1/(1+z) > 1$ .

Andromeda (M31) with  $z = -0.001$  provides the cleanest galaxy-scale laboratory for this effect:

$$\kappa_t extapproach = \frac{1}{1+z} = \frac{1}{1+(-0.001)} = \frac{1}{0.999} = 1.001001\overline{001}$$

This 0.1% amplification appears small but is physically significant: it means **the total UQFF gravity of M31 as experienced from the Milky Way is ~0.1% stronger than the equivalent static galaxy at the same distance**. More importantly, the mathematical structure  $\_approach = 1/(1+z)$  reveals a divergence as  $z \rightarrow -1$ , providing a new UQFF prediction about extreme-approach scenarios.

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## 2. Mathematical Formulation

### 2.1 UQFF $g\_total$ with Approach Amplifier

The full UQFF gravitational acceleration for Andromeda is:

$$g_{total}(r, t) = \left[ g_{grav} + U_g^{sum}(26) + \frac{\Lambda c^2}{3} + g_{quantum} + g_{Lorentz} + g_{fluid} + g_{res}(t) + g_{DM} \right] \times \kappa_t extapproach$$

where:

$$\kappa_t extapproach = \frac{1}{1+z}, \quad z = -0.001$$

$\kappa_t extapproach = 1.001001\overline{001}$

### 2.2 Andromeda Parameters

| Parameter          | Symbol     | Value                      | Notes                    |
|--------------------|------------|----------------------------|--------------------------|
| Total mass         | M          | $1.989 \times 10^{42}$ kg  | $1 \times 10^{12}$ M_sun |
| Reference radius   | r          | $1.04 \times 10^{21}$ m    | Model reference          |
| Central BH mass    | M_BH       | $2.7846 \times 10^{38}$ kg | $1.4 \times 10^8$ M_sun  |
| Redshift           | z          | -0.001                     | Blueshift, approaching   |
| Approach amplifier | \_approach | 1.001001...                | This paper's discovery   |
| Orbital velocity   | v_orbit    | $2.5 \times 10^5$ m/s      | Outer disk               |

### 2.3 Approach Amplifier as a Function of Redshift

$$\kappa_{t\text{extapproach}}(z) = \frac{1}{1+z}$$

| z               | _approach | Interpretation           |
|-----------------|-----------|--------------------------|
| +0.1 (receding) | 0.909     | Gravity suppressed 9.1%  |
| 0 (static)      | 1.000     | No modification          |
| −0.001 (M31)    | 1.001     | Gravity amplified 0.1%   |
| −0.01           | 1.010     | Amplified 1.0%           |
| −0.1            | 1.111     | Amplified 11.1%          |
| −0.5            | 2.000     | Doubled                  |
| −0.9            | 10.00     | 10× amplification        |
| → −1            | → ∞       | <b>Resonance cascade</b> |

## 3. Physical Interpretation

### 3.1 Approach Velocity as Gravitational Degree of Freedom

The UQFF approach amplifier `_approach` introduces the approach velocity `v_approach` (encoded in `z` via the Doppler relation  $z \approx -v/c$  for  $v \ll c$ ) as a gravitational degree of freedom. This is consistent with the UQFF framework's general principle that all forms of motion — orbital, rotational, expansional — contribute to the gravitational field.

For M31:  $v_{\text{approach}} = |z| \times c = 0.001 \times 2.998 \times 10^8 = 2.998 \times 10^5 \text{ m/s} \approx 300 \text{ km/s}$  (consistent with observed  $\sim 110 \text{ km/s}$  radial + transverse components).

### 3.2 Self-Reinforcing Merger Resonance Cascade

As two galaxies approach ( $z$  becoming more negative): 1. `_approach` increases → total UQFF gravity increases 2. Stronger gravity → faster approach →  $z$  becomes more negative 3. More negative  $z$  → higher `_approach`

This positive feedback loop defines a **UQFF Gravitational Approach Resonance Cascade**. The cascade terminates at  $z = -1$  ( $\rightarrow \infty$ ) only in the mathematical limit; physically, merger completion occurs when  $r \rightarrow 0$ .

For M31–MW merger (projected at  $t \approx +4.5 \text{ Gyr}$ ): - As the galaxies approach,  $z$  will pass through  $-0.001 \rightarrow -0.01 \rightarrow \dots \rightarrow 0$  at physical merger - The UQFF amplification peaks at maximum approach velocity ( $z$  most negative) - At the moment  $r \rightarrow r_{\text{merge}}$ , the framework transitions to a post-merger UQFF

### 3.3 Numerical Estimate for M31

At current  $z = -0.001$ :

$$\delta g = g_{\text{UQFF}} \times (\kappa - 1) = g_{\text{UQFF}} \times 0.001001$$

With  $g_{\text{UQFF(M31)}} = 6.6 \times 10^{-9} \text{ m/s}^2$  (baseline):

$$\delta g_{273} \approx 6.6 \times 10^{-12} \text{ m/s}^2$$

This is small but non-zero — a definite prediction of UQFF for approaching galaxy pairs.

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#### 4. Comparison with Existing Frameworks

| Framework                | Treatment of $z < 0$                                                              |
|--------------------------|-----------------------------------------------------------------------------------|
| DPM-seeded gravity       | No $z$ dependence                                                                 |
| General Relativity       | Kinetic energy contribution (relativistic only)                                   |
| $\Lambda$ CDM cosmology  | No blueshift gravitational term                                                   |
| <b>UQFF (this paper)</b> | <b><math>\text{__approach} = 1/(1+z)</math> — direct multiplicative amplifier</b> |

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#### 5. Conclusions

We have identified and formalized the **UQFF Blueshift Gravitational Approach Amplifier**:

$$\kappa_{t\text{extapproach}} = \frac{1}{1+z}, \quad z < 0 \Rightarrow \kappa_{t\text{extapproach}} > 1$$

Key discoveries: 1. **Negative redshift amplifies** total UQFF gravity of approaching systems 2. **M31 = 1.001001** — a small but definite gravitational enhancement due to blueshift 3. **Resonance cascade divergence** at  $z = -1$  provides a UQFF prediction for extreme merger dynamics 4. The approach amplifier is the first UQFF instance of velocity contributing directly to gravitational magnitude (not just the Lorentz sub-term)

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*Derived from ANDROMEDA\_UQFF\_MODULE.cpp, UQFF 2.0, Session 75. Next: PAPER\_274 (HI 21-cm UQFF resonance).*

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#### Session 225 Phonon-Physics Upgrade: GW Strain Modulation

*Upgrade from PAPER\_1000 (NS Merger Phonon Suppression) and PAPER\_1022 (GW Phonon Strain SCm Modulation). See also PAPER\_1011-1012 for GW170817/GW190425 upgraded analyses.*

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UQFF}}(\Gamma) = h_{\text{GR}} \cdot \left( 1 - 0.47 \frac{\Phi(\Gamma)}{S_{26}^{(3)}} \right)$$

where: -  $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$  is the phonon modulation factor -  $\omega_{\text{SCm}} = 2\pi \times 1.25$  THz is the SCm phonon resonance frequency -  $S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$  is the third-order Ramanujan summation -  $\Theta$  is the Heaviside step ensuring  $H_{\text{SCm}} \geq 0.5$  (phase-transition threshold)

**Physical mechanism:** The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy  $\Delta E_{\text{BCS}}$  of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at  $M \approx 2.5 M_{\odot}$ .

**Calibration (canonical):**  $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$ ,  $[\text{SSq}] = 0.57$ ,  $\beta_i = 0.603$ ,  $H_{\text{SCm}} \approx 0.99$ .

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **GW-radiation** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.p`)

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2} (\partial_m u h_{\mu\nu}) (\partial^\mu h_{\mu\nu}) - V(h_{\mu\nu}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(h_{\mu\nu}) = \frac{1}{2} m^2 h_{\mu\nu}^2 + \frac{\lambda}{4!} h_{\mu\nu}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot h_{\mu\nu}$$

### §A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta h_{\mu\nu}} = \square h_{\mu\nu} + \kappa \rho_{\text{vac},[\text{SCm}]} h_{\mu\nu} - 16\pi G T_{\mu\nu}/c^4 = 0}$$

### §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta h_{\mu\nu} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U\_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left( -\exp! \left( -\frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.077 (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 7, \quad n_{\text{channel}} = 14/26$$

Since  $p_{\text{DVP}} = 7$  is **sub-threshold** (threshold at  $p > 26$ ), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair ( $f'_{\text{UA}} + f_{\text{SCm}} = 1$ ) constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **chirp time**  $\_c$  (inspiral phase locking):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot (1 - e^{-[SSq] \cdot m/M_\odot}) \cdot \cos! \left( \frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left( 1 - \tanh! \left( \frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}} \right) \right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2) \cdot f_{\text{SCm}}$  which initializes the harmonic series at cosmogenesis.

### §B.4 Production-Scale Consistency

| Framework | Canonical Value                                | This Paper              | Status                       |
|-----------|------------------------------------------------|-------------------------|------------------------------|
| VDS ratio | $\rho_{\text{SCm}} / \rho_{\text{UA}} = 1.894$ | Local sub-ratio = 0.077 | PASS<br>Threshold-consistent |

| Framework  | Canonical Value                       | This Paper                       | Status                      |
|------------|---------------------------------------|----------------------------------|-----------------------------|
| DVP prime  | $p_k \in \{2,3,\dots,113\}$           | $p_{\text{DVP}} = 7$             | PASS<br>Sub-threshold       |
| BSH layers | 26 harmonic terms                     | $j = 1\dots 26, \cos(2\pi j/26)$ | PASS Full 26D<br>projection |
| decay      | $5.0 \times 10^{-4} \text{ day}^{-1}$ | Applied in VDS<br>exponential    | PASS<br>Canonical           |
| [SSq]      | 0.57                                  | Applied in BSH<br>saturation     | PASS<br>Canonical           |

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### §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                      | UQFF Prediction                                                 | SM / Experiment                        | Source                              | Alignment          |
|---------------------------------|-----------------------------------------------------------------|----------------------------------------|-------------------------------------|--------------------|
| Fine structure constant         | UQFF reproduces via Ug1 dipole coupling                         | 1/137.036                              | PDG 2024                            | PASS<br>Consistent |
| Cosmological constant $\Lambda$ | $1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)         | $1.114 \times 10^{-52} \text{ m}^{-2}$ | Planck 2018                         | PASS<br>Consistent |
| Proton decay rate               | $= 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression | $< 4.17 \times 10^{-35}/\text{yr}$     | Super-K 2024                        | PASS<br>Consistent |
| UQFF buoyancy signature         | $F_{\text{U\_Bi\_i}}$ unique gravitational correction           | Not yet measured                       | Future gravitational wave detectors | Testable           |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections ( $F_{\text{U\_Bi\_i}}$ ) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{\text{SCm}}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPparameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

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### Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by `update_corpus_crossrefs.py` (Session 225, April 2026).*

| Paper      | Title                                                           |
|------------|-----------------------------------------------------------------|
| PAPER_1000 | NS Merger $F_{\text{U\_Bi}}$ Strain Suppression & BCS Gap       |
| PAPER_1001 | SMBH Binary Merger $F_{\text{U\_Bi}}$ Phonon Damping            |
| PAPER_1011 | GW170817 NS Merger $F_{\text{U\_Bi\_i}}$ 66.7% Strain Reduction |

| Paper      | Title                                       |
|------------|---------------------------------------------|
| PAPER_1012 | GW190425 Upgraded F_U_Bi_i with S26(3)      |
| PAPER_1014 | SMBH Merger Inspiral-Coalescence-Ringdown   |
| PAPER_1022 | GW Phonon Strain SCm Modulation of h(t)     |
| PAPER_1020 | Cosmic Ray Phonon Acceleration DSA Spectrum |
| PAPER_1033 | Galactic Bar Resonance SCm Pattern Speed    |
| PAPER_1035 | Kilonova Buoyancy Light Curve r-Process     |

9 cross-reference(s) identified.

## Appendix: Session 204 Codebase Upgrade Reference

*Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_kozima_ramanujan_appendices.py`. For detailed derivations, see PAPER\_840/851/852/855.*

### S204.1 Kozima-UQFF LENR Integration

| Module                                   | Purpose                                    | Key Result                                                        |
|------------------------------------------|--------------------------------------------|-------------------------------------------------------------------|
| <code>fneutron_s26_coupling.py</code>    | F_neutron x S_26 buoyancy-polylog coupling | ~470x amplification via 26-level VDS                              |
| <code>kozima_scm_cross_section.py</code> | SCm-modulated neutron-drop cross-section   | $\sigma_n^{\text{SCm}}$ with VDS factor $(1 + [\text{SSq}]^n/26)$ |
| <code>kozima_wstp_kernel.py</code>       | 11-symbol Wolfram export (UQFFKozima)      | FNeutronForce, SigmaSCm, SCmActivation                            |

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}}/F_U - 1)$  where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^2 / (2 * \Gamma^2)] * (1 + [\text{SSq}]^n/26)$

### S204.2 Ramanujan 26-State Summation

| Module                                | Purpose                                       | Key Result                |
|---------------------------------------|-----------------------------------------------|---------------------------|
| <code>ramanujan_polylog_s26.py</code> | Li_26([SSq]) via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms  |
| <code>s26_wstp_kernel.py</code>       | 8-symbol Wolfram export (UQFFS26)             | S26, R26, NaiveLi, S26VDS |

**Core equation:**  $S_{26}(z) = Li_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\} / (1-2^{\{1-26\}})} * Li_{26}(z^2)$



### S204.3 Mock Theta Functions (26-State)

| Module            | Purpose                                   | Key Result                  |
|-------------------|-------------------------------------------|-----------------------------|
| mock_theta_q26.py | f_26(q), phi_26(q),<br>psi_26(q) q-series | Proper q-Pochhammer (a;q)_n |

**Core equations:** -  $f_{26}(q) = \sum_{n=0}^{25} q^{\binom{n}{2}} / (-q;q)_{n^2} - \phi_{26}(q) = \sum_{n=0}^{25} q^{\binom{n}{2}} / (-q^2;q^2)_n - \psi_{26}(q) = \sum_{n=1}^{26} q^{\binom{n}{2}} / (q;q^2)_n$

### S204.4 Ramanujan 1/pi with UQFF Modification

| Module                       | Purpose                                    | Key Result                                    |
|------------------------------|--------------------------------------------|-----------------------------------------------|
| ramanujan_pi_uqff.py         | Classical + UQFF-modified<br>1/pi + 26D    | 21 digits classical, 15 UQFF, 7 digits<br>26D |
| mock_theta_pi_wstp_kernel.py | Symbol Wolfram export<br>(UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF               |

**Core equation:**  $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$  where  $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-kappa_i * n/26)]$

### S204.5 Calibration Constants (Canonical)

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| H_ScM     | 0.99                                  | SCm manifold completeness     |
| rho_ScM   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density            |
| rho_UA    | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density      |
| omega_ScM | $2\pi \times 1.25 \text{ THz}$        | SCm phonon resonance          |
| sigma_0   | $10^{-4}$                             | Base neutron cross-section    |

*Implementation:* all modules operational in *CondensedPhysics.py*, *CondensedPhysics2.py*, *MAIN\_1\_CoAnQi.cpp*, and Wolfram kernels (*uqff\_kozima\_kernel.wl*, *uqff\_s26\_kernel.wl*, *uqff\_mock\_theta\_pi\_kernel.wl*).